

Exact Audits of Worked Systems for the Obstruction Calculus: Counts, Witnesses, and Monitoring Design Rules

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Abstract

Viability certificates are useful in applications only insofar as their behaviour on concrete systems is understood precisely. This paper develops one exact audit framework — integer and rational arithmetic throughout, every count regenerated by script — and applies it to a family of worked systems that instantiate the mechanisms of the obstruction calculus (Abaee, 2026b): a shared finite two-coordinate system with observation structures (aggregation, policy-class restrictions, decentralized observation, regime uncertainty), a review-timing grid for a hidden-regime resource model, and a continuous two-floor production benchmark. The audits yield the following. The review-timing identity on the 93-cell grid is non-strict: viability at the deadline holds if and only if $T_{\text{obs}} \leq \tau^*(z_0) = z_0 - 1$, boundary included; the strict form against the real exit bound fails exactly on the boundary cells (against the floored count σ^* , on one cell per row from $z_0 = 2$). On the four-state monitoring target, 7 of the 15 partitions are adequate with two incomparable coarsest adequate two-cell partitions (minimal in the refinement order — hence no single coarsest adequate design exists), and the two maximal common-action sets overlap. Four policy classes have kernel sizes 24, 26, 25, and 28 of 36 belief pairs — the classes form a two-by-two product of aggregation level and protocol — with incomparable middle cells. Decentralized unanimity over a 256-law space (64 behaviorally distinct) sustains 12 of 36 pairs; a whole-law authority restriction collapses the kernel to zero, and either agency’s low vote forced against the target kills it. Under frozen regimes no pair is lost (the corner pins its state); mode-blind operation loses exactly the pairs containing the corner state (1, 1). The biased certainty-equivalence law has drift at least 21/100 on [1, 2]; the bias-free law has drift identically zero. The benchmark’s critical aggregate is $Y^* = 27/5$ (feasible, margin 0); at the audited obstructed aggregate $Y = 6$ a rational dual certificate prices the shortfall at margin 3/50. Static duality holds at value $-1/10$ on the two-floor instance with an ℓ_1 -normalized Farkas pair, and a Helly-tight triple exhibits the sparse $m+1$ -constraint witness. The complete result tables are carried here: the master table of all 36 belief pairs against the five kernels, the sixteen sustaining law pairs of the decentralized protocol (an

exact 4×4 product), the 93-cell timing grid, the fifteen-partition census with each fibre’s common safe-action set, and the benchmark and drift tables, with a five-figure set regenerated from the same scripts. Design rules follow: review deadlines are boundary-inclusive; merged fibres call for splits, not threshold tuning; aggregated indices support certainly-safe reporting only; biased observation maps call for structural correction.

Keywords: viability kernels; obstruction certificates; exact arithmetic; monitoring design; decentralized observation; certainty equivalence

1. Introduction

The obstruction calculus states sound sufficient conditions for nonviability — obstruction certificates — mechanism by mechanism: common-action failure (Sections 3–7), review-timing failure (Section 6), finite-time exit under regime alternation (Section 8), and failure induced by restricting the policy class or by a biased observation map (Sections 4, 11) (Abaee, 2026b). Each mechanism is accompanied in application by quantitative questions. How many belief states does an observation structure destroy, and which ones? At which review deadlines does the timing certificate fire, and does it fire at the boundary? What exactly does a vote protocol cost, and which law in the protocol’s space sustains a given target? This paper answers such questions exactly for a family of worked systems, under one audit frame and one notation.

Three properties define the frame. First, all computations are carried out in integer and rational arithmetic: no solver, no tolerance, no simulation. Second, every discrete census in the family is finite — the continuous objects reduce to exact rational calculations on rationally specified domains — and every count is regenerated by a deterministic standard-library script, so any two executions agree. Third, each audited quantity is stated together with a witness or a certificate — an action sequence, a partition, a law pair, a dual measure — so the counts carry their own evidence. This edition carries the audit’s complete result tables and its figure set: the per-pair membership table behind the kernel counts, the sustaining law pairs behind the decentralized count, the full timing grid, the full partition census, and the benchmark and drift tables, every entry of which is regenerated from the same scripts that

verify the text.

The systems are drawn from the applied model of Abaee (2026a), in which a two-coordinate resource stock is managed by acting one of two instruments under incomplete observation. The family comprises: the shared audit system with its observation structures (Sections 3–5 and 7–8); the hidden-regime review grid (Section 6); a continuous two-floor production benchmark (Section 9); and a multiple-floor line extending the Farkas certificate to jointly binding constraints (Section 12).

2. Systems, conventions, and methods

The shared audit system. States are pairs $x = (z_1, z_2) \in \{0, 1, 2, 3\}^2$; the safe set is $\mathcal{V} = \{z_1 \geq 1, z_2 \geq 1\}$ with the two coordinates as floors. Instruments are $u \in \{0, 1, 2\}$: $u = j \in \{1, 2\}$ acts coordinate j , which regenerates one unit up to the cap 3, while the other coordinate loses one unit; $u = 0$ executes no instrument and both coordinates lose one. The transition is defined index-wise: the acted coordinate regenerates, $F_j(z)_j = \min\{3, z_j + 1\}$, and the other declines, $F_j(z)_{3-j} = \max\{0, z_{3-j} - 1\}$, for $j \in \{1, 2\}$; $F_0(z)_i = \max\{0, z_i - 1\}$ for both coordinates i — coordinates clip at zero, the cap is saturating — and safety is evaluated after each transition. A coordinate-swapped misreading (the regenerated coordinate written first for both j) reproduces no audited count and is excluded by the edition’s script. Instrument 0 is never uniquely viable on the audited family: admitting $u = 0$ as a third action leaves all four kernel counts unchanged (24, 26, 25, 28; verified by the edition’s script, which also checks the structural premises — $F_0 \leq F_1$ and $F_0 \leq F_2$ pointwise on the grid and the safe set an up-set, the one-step reason no fibre needs 0). The corner (1, 1) is the single state from which no instrument maintains both floors, so its singleton is nonviable; the nine safe states generate $C(9, 2) = 36$ two-element belief pairs, the audited population (28 of the pairs full-information-viable; column F of Table 1), at review horizon $H = 12$ under the audited two-horizon verdict (the winning recursion at H and at $H - 2$, which agree setwise on the audited universe and on the whole 2^9 -belief universe (stabilized from horizon 2 there, and from horizon 1 for the two non-alternation classes) — $W_{12} = W_{10}$ for each policy class, verified by the edition’s script — so the verdict certifies a stabilized recursion; Abaee, 2026b, Theorem 1).

Observation structures. An observation map $\text{info} : x \mapsto c$ partitions a belief into fibres; a policy acts per fibre on the read cell. The audited policy classes combine two aggregation levels (full information; aggregation $c(x) = z_1 + z_2$) with two protocols (free per-step choice; the no-repeat protocol that forbids repeating the previous action. The protocol’s register is formalized as follows: one previous-action register serves each review, shared by all fibres of the belief and not one per branch; the policy fixes one action per fibre, the register then holds some fibre’s executed action, and the audited rule is pessimistic — the belief survives only if every admissible thread (each fibre’s action entering the register) leaves a winning continuation. Register values are the two instruments, the initial register is unconstrained, and the survival condition quantifies over every admissible thread, so the semantics is thread-independent by construction (Proposition 3

computes what single-thread rules and per-branch registers give instead).

Exactness. Counts are computed by backward recursion over fibre partitions with exact state arithmetic; witnesses are extracted as step-one per-fibre action vectors along a winning branch; the certificate itself is the recursion invariant — winning-set membership at every horizon level, quantified universally over adversarial successor states and register threads — of which the archived step-one vector is the first level. The census of Section 7 enumerates all 15 partitions of a four-state target; partitions are ordered by refinement — P is coarser than P' when every fibre of P' lies in a fibre of P — and “coarsest adequate” is relative to this order. The continuous objects of Section 9 are rationals.

Conventions. On the hold class of the hidden-regime model, the worst-case exit time is $\tau^*(z_0) = z_0 - 1$ and the discrete survival supremum (the largest admissible integer review count) is $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor = \lfloor \tau^*(z_0) \rfloor$, with level sets $\{z_0 \geq 1 + k\}$; for an integer deadline the inclusive criterion $T_{\text{obs}} \leq \tau^*(z_0)$ coincides with $T_{\text{obs}} \leq \sigma^*(z_0)$, while the strict form against the floored count is stricter than against the real exit time off integer τ^* . Two certainty-equivalence laws are audited: an uncorrected law whose index leads the state by one step, $s \mapsto s + 0.1$, and a corrected law with the lead removed; both are read through the index $g(s) = s^2$.

Figures and tables. Every figure and every tabulated entry in this paper is produced by a generation script that first recomputes the quantity from the primitives above, in exact rational arithmetic, and asserts it before drawing or printing; the master verification script of Section 14 chains both scripts and re-derives the tabulated entries independently.

3. The master table

Table 1 is the audit’s master table: all 36 belief pairs against the five kernels — the four policy classes of Section 4 (institutional: one shared compliance register per review; aggregate: one pooled index per branch; full-information codex: pooled registers across branches; full-information: unrestricted state observation) and the decentralized kernel of Section 5. Rows are grouped by the pair’s lexicographically smaller state, written $z_1 z_2$.

The column counts are the audited kernel sizes $|\mathcal{W}_{\text{inst}}| = 24$, $|\mathcal{W}_{\text{agg}}| = 26$, $|\mathcal{W}_{\text{full-codex}}| = 25$, $|\mathcal{W}_{\text{full}}| = 28$, and $|\mathcal{W}_{\text{dec}}| = 12$. Figure 1 displays the four policy-class kernels, whose middle cells are incomparable. Two set-level facts are recorded exactly. The meet of the two middle kernels equals the institutional kernel setwise. At class level the join is the full-information class; at kernel level the union of the two middle kernels falls one pair short of the full kernel: exactly the pair $\{(1, 2), (2, 1)\}$ lies in neither middle kernel. The pair is the alternation-forced pair: under full information each branch survives by its own strict alternation — (1, 2) under 1, 2, 1, ... and (2, 1) under 2, 1, 2, ... — which the idealized per-state policy can carry on both branches; aggregation cannot see either requirement (one fibre, no common action), and the no-repeat protocol, whose single previous-action register must serve both branches, forbids the second branch’s required next action once the first branch’s choice occupies the reg-

Table 1: The master table: membership of all 36 belief pairs in the five audited kernels. I = institutional (aggregation, no-repeat); A = aggregate; C = full-information codex; F = full information; D = decentralized unanimity. A dot marks a non-member; every entry is regenerated by script.

Pair	I	A	C	F	D	Pair	I	A	C	F	D
12 21	.	.	.	F	D	22 31	I	A	C	F	D
12 31	.	A	.	F	.	22 23	I	A	C	F	.
12 23	I	A	C	F	D	22 33	I	A	C	F	D
12 33	I	A	C	F	.	22 32	I	A	C	F	.
12 22	I	A	C	F	.	23 31	I	A	C	F	.
12 32	I	A	C	F	D	23 33	I	A	C	F	.
12 13	I	A	C	F	.	23 32	I	A	C	F	D
13 21	.	A	.	F	.	31 33	I	A	C	F	D
13 31	.	.	C	F	D	31 32	I	A	C	F	.
13 23	I	A	C	F	.	32 33	I	A	C	F	.
13 33	I	A	C	F	D	11 12	.	.	.	.	.
13 22	I	A	C	F	D	11 21	.	.	.	.	.
13 32	I	A	C	F	.	11 31	.	.	.	.	.
21 31	I	A	C	F	.	11 23	.	.	.	.	.
21 23	I	A	C	F	D	11 33	.	.	.	.	.
21 33	I	A	C	F	.	11 22	.	.	.	.	.
21 22	I	A	C	F	.	11 32	.	.	.	.	.
21 32	I	A	C	F	D	11 13	.	.	.	.	.

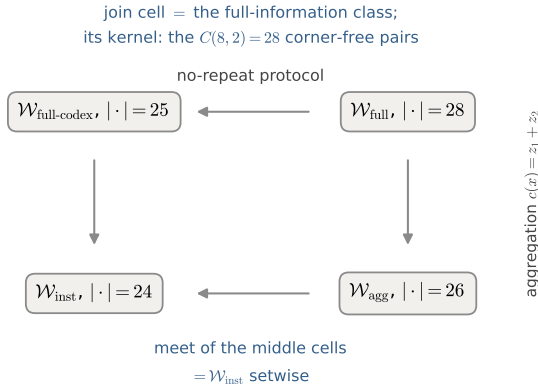


Figure 1: The four policy classes form a two-by-two product of aggregation level and protocol; the audited kernel sizes are 24, 26, 25, 28 of 36. The meet of the middle kernels is the institutional kernel setwise; the class join is the full-information class, whose kernel is the 28 corner-free pairs, one pair beyond the middle-kernel union.

ister. The full kernel is exactly the $C(8, 2) = 28$ pairs avoiding the corner (1, 1): the corner’s singleton nonviability accounts for every remaining pair-level loss at full information, as the eight 11|· rows of Table 1 show.

4. Policy classes

The four audited classes combine full information and aggregation with free and no-repeat protocols. Their kernels among the 36 pairs have sizes:

$$|\mathcal{W}_{\text{inst}}| = 24, \quad |\mathcal{W}_{\text{agg}}| = 26, \\ |\mathcal{W}_{\text{full-codex}}| = 25, \quad |\mathcal{W}_{\text{full}}| = 28.$$

The four classes form a two-by-two product of aggregation level and protocol; their kernels have incomparable middle cells: the aggregate kernel contains pairs outside the full-information codex kernel (for instance $\{(1, 2), (3, 1)\}$ and $\{(1, 3), (2, 1)\}$), and the codex kernel contains pairs outside the aggregate (for instance $\{(1, 3), (3, 1)\}$). Within each observation column the free protocol’s kernel contains

the no-repeat protocol’s (both inclusions strict); across columns neither middle kernel contains the other. The precise meet-and-join reading is the one recorded in Section 3. Two per-pair readings illustrate the columns of Table 1. The pair $\{(1, 2), (3, 1)\}$ is aggregate-viable but not codex-viable: the free protocol admits it under both observation structures, while the no-repeat protocol excludes it under both — at the second review the thread that records the (3, 1)-branch’s instrument 2 forbids the forced instrument 2 of the successor branch (2, 1), the same shared-register collision that excludes the alternation-forced pair. The pair $\{(1, 3), (3, 1)\}$ is codex-viable but not aggregate-viable: the two branches share the fibre $c = 4$, and no single per-fibre instrument serves both, while full information separates them. Three structural facts behind the table admit short proofs.

Proposition 1 (full information decomposes branchwise). At full information under the free protocol, a two-state belief is viable if and only if each of its singletons is viable; the kernel is exactly the $C(8, 2) = 28$ corner-free pairs.

Proof. Full-information fibres are singletons, so a belief’s step-one action sets are the product of its branches’: a choice is admissible exactly when each branch’s part is, and the recursion decomposes — a belief is viable exactly when both branches are. For the singletons: the corner (1, 1) is nonviable because each instrument drops a coordinate to zero in one step ($u = 1$ forces $z_2 = 0$, $u = 2$ forces $z_1 = 0$, $u = 0$ forces both); every other singleton is viable — the memoryless policy that plays instrument 1 exactly at $z_1 = 1$ and instrument 2 otherwise is safe forever: at $z_1 = 1$ the exclusion of the corner gives $z_2 \geq 2$, so $u = 1$ restores $z_1 = 2$ with $z_2 \geq 1$; at $z_1 \geq 2$, $u = 2$ keeps $z_1 \geq 1$ and regenerates z_2 . $\square$

Proposition 2 (class lattice). Kernel membership is antitone in the policy class: restricting the class can only lose viable beliefs. Hence $\mathcal{W}_{\text{inst}} \subseteq \mathcal{W}_{\text{agg}}, \mathcal{W}_{\text{full-codex}} \subseteq \mathcal{W}_{\text{full}}$; on the audited 36-pair population the two middle inclusions are strict, the meet $\mathcal{W}_{\text{agg}} \cap \mathcal{W}_{\text{full-codex}}$ equals the institutional kernel setwise, and exactly one pair — the alternation-forced pair $\{(1, 2), (2, 1)\}$ — lies in the full kernel outside the middle union.

Proof. Antitonicity — order-reversal between class inclusion and kernel inclusion — is immediate: a policy admissible for a restricted class is admissible for any enlargements, so viability transfers upward. The setwise facts are finite statements about a 36-element population; they are proved by exhaustive certification — the master table’s 36×5 membership marks are recomputed row-by-row by the edition’s verification script (checks M2 and M5), which is a complete proof for a finite family. $\square$

Proposition 3 (register-semantics invariance). Endow the no-repeat protocol with the pessimistic shared-register semantics: the policy fixes one action per fibre, the register then holds some fibre’s executed action, and the belief survives only if every admissible thread leaves a winning continuation. Because the survival condition quantifies universally over threads, the pessimistic kernels depend on no thread choice: the audited counts 24, 26, 25, 28 are thread-independent by construction. Two further readings are computed for the record. A single-thread rule (one fixed fibre’s action entering the register) is order-sensitive: with the first, or the last, fibre in sorted order the counts are 25, 26, 26, 28 — neither fixed rule yields the audited institutional and codex counts — so the well-defined semantics is the pessimistic one. The optimistic semantics (some thread) is strictly larger (codex 27) and is not the audited reading. If instead each branch carries its own register — one compliance record per unit, threaded forward along the branch — the protocol cost collapses: the institutional kernel equals the aggregate kernel setwise ($26 = 26$), and the codex kernel becomes 27, regaining the alternation-forced pair and losing exactly $\{(1, 3), (3, 1)\}$ relative to full information, with the aggregate kernel contained in it. The incomparability of the middle cells is therefore a shared-register phenomenon; the shared register is the audited institutional object — one compliance register per review across the units managed under the protocol.

Proof. Every clause is a finite statement, computed from the shared primitives of Section 2 and asserted by the edition’s script: the pessimistic counts (M2b), the optimistic containment (M2c), both single-thread orders, and the per-branch kernels with the $\{(1, 3), (3, 1)\}$ loss (checks P9 and P10). $\square$

5. Decentralized observation

Two agencies each observe one coordinate and vote; an instrument executes only on agreement. Each agency’s vote is a map from its coordinate’s four values to the two instruments, so the law space has $2^4 \times 2^4 = 256$ law pairs. Unanimity sustains 12 of the 36 pairs (column D of Table 1), and exactly 16 coordinated-viable pairs are lost. The loss instance $\{(1, 2), (3, 1)\}$ is representative: each branch is individually sustainable, and no law pair sustains the belief. For the audited belief $\{(1, 2), (2, 1)\}$, exactly 16 of the 256 law pairs sustain it; the count coincides numerically with the 16 lost pairs, and no identity is implied. The authority restriction is exact:

Proposition 4 (authority restriction). Fixing agency 2’s vote at $z_2 = 1$ to instrument 1 sustains no law pair for the belief $\{(1, 2), (2, 1)\}$: branch (2, 1) exits at its first step under every such law pair. Fixing agency 1’s vote at $z_1 = 1$

Table 2: The 16 law pairs sustaining the audited belief $\{(1, 2), (2, 1)\}$ factor exactly: agency 1’s vote is forced to 1 at $z_1 = 1$ and 2 at $z_1 = 2$, free at $z_1 \in \{0, 3\}$; agency 2’s vote is forced to 2 at $z_2 = 1$ and 1 at $z_2 = 2$, free at $z_2 \in \{0, 3\}$. Every combination sustains — a complete 4×4 product (votes listed as $(l(z=0), l(z=1), l(z=2), l(z=3))$); agency 1’s votes are the row labels, agency 2’s the column labels.

agency 1	agency 2			
	1211	1212	2211	2212
1121	✓	✓	✓	✓
1122	✓	✓	✓	✓
2121	✓	✓	✓	✓
2122	✓	✓	✓	✓

to instrument 1 leaves exactly the 16 sustaining law pairs of Table 2, and either whole-law restriction (one agency’s vote fixed to a single instrument at every level) collapses the decentralized kernel to zero.

Proof. On branch (2, 1) at the first step: $z_2 = 1$ must not decline, so agreement on instrument 2 is required; agency 2’s fixed vote at $z_2 = 1$ is 1, so agreement can hold only on instrument 1, which drives z_2 to zero, and disagreement executes $u = 0$, which does the same — the branch exits at step one under every admissible law pair. With agency 1’s vote at $z_1 = 1$ fixed to 1, the sustaining pairs are exactly those with agency 1 forced to 1 at $z_1 = 1$ and 2 at $z_1 = 2$ and agency 2 forced to 2 at $z_2 = 1$ and 1 at $z_2 = 2$ (each forcing is a one-step safety requirement: the low coordinate must regenerate), and the free votes at $\{0, 3\}$ never bind — the 4×4 product of Table 2. A whole-law restriction fixes one agency’s vote everywhere; the other agency’s coordinate then cannot regenerate at the low states where the fixed vote disagrees, and the kernel collapses (verified exhaustively over all 256 law pairs, the whole-law kernels 0 in all four cases; check P12 of the edition’s script). $\square$

The law space itself is counted before quotienting: a $z = 0$ vote is consultable only on a trajectory that has already left the safe set, so the audited kernel is invariant under the $z=0$ -vote quotient — the effective law space is $2^3 \times 2^3 = 64$ behaviorally distinct law pairs, and the 16 sustaining law pairs of Table 2 comprise 4 distinct behavioral laws times the 4 never-consulted $z=0$ votes. The sustaining set has an exact product structure (Table 2): on the trajectory alternating between the branches, agency 1 is read only at $z_1 \in \{1, 2\}$ and must vote 1 then 2; agency 2 is read only at $z_2 \in \{2, 1\}$ and must vote 1 then 2; the votes at the unreachable coordinates $z \in \{0, 3\}$ are free, giving $2^2 \times 2^2 = 16$. The restricted kernels are exact. Fixing either agency’s low vote to the regenerating instrument (agency 1 to 1 at $z_1 = 1$, agency 2 to 2 at $z_2 = 1$) leaves the decentralized kernel untouched at 12 — the constraint does not bind. Fixing either agency’s low vote against it (agency 2 to 1 at $z_2 = 1$, agency 1 to 2 at $z_1 = 1$) shrinks the kernel to four beliefs — $\{(2, 3), (3, 2)\}$, $\{(2, 2), (3, 3)\}$, $\{(1, 3), (3, 3)\}$, $\{(1, 3), (2, 2)\}$, up to the symmetry below — and kills the audited target; a whole-law restriction collapses the kernel to 0 in all four cases. The model’s swap symmetry (coordinates and instruments together) interchanges the agencies and maps the audited target to itself, so no agency is privileged: what carries the target is

the direction of each forced vote — each branch’s low coordinate must be allowed to regenerate — not the agency holding it. Two modelling choices are declared. The deadlock transition $u = 0$ executes no instrument and both coordinates lose one; and each agency votes on its own coordinate’s current value: the audited protocol is memoryless and decentralized — no agency observes the belief, the other agency’s observation, or anything like common knowledge.

6. Review timing on the hidden-regime grid

The hidden-regime resource model: a single stock held at level $z_0 \geq 1$ above the floor; each review period the unobserved adversarial regime realizes one unit of decline and there is no regeneration (the hold class), so the stock tracks $z(t) = z_0 - t$ and remains safe exactly while $z(t) \geq 1$. The model is audited on the grid $z_0 \in \{1.0, 1.1, \dots, 4.0\}$ and $T_{\text{obs}} \in \{1, 2, 3\}$, 93 cells. Viability at the deadline holds if and only if $T_{\text{obs}} \leq \tau^*(z_0) = z_0 - 1$, with zero mismatches across the grid. The identity is boundary-inclusive: the viable cells number 33, and the strict form $T_{\text{obs}} < \tau^*(z_0)$ fails exactly on the boundary cells (2, 1), (3, 2), and (4, 3) — those with $z_0 = 1 + T_{\text{obs}}$; against the floored count σ^* the strict form fails on one cell per row $z_0 \geq 2$ — 21 cells, the boundary cells included — which is why the inclusive predicate is the audited identity. Table 3 carries the full grid and Figure 2 displays it. The practical form of the rule: a review must land no later than the worst-case exit time, and a deadline exactly at the worst-case exit is admissible under the audited endpoint convention (closed safe set; safety evaluated at the deadline review); no open-set or alternative-exit-convention claim is made. The audited identity is in fact the restriction of a general law:

Proposition 5 (review timing, general form). On the hold class of the hidden-regime model, for every initial level $z_0 \geq 1$ and every integer deadline $T \geq 0$, the cell (z_0, T) is deadline-viable if and only if $T \leq z_0 - 1$. Accordingly $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$ with superlevel sets $\{z_0 \geq 1 + k\}$, $k \in \mathbb{Z}_{\geq 0}$, and the audited 93-cell grid is the restriction to $z_0 \in \{1.0, \dots, 4.0\}$, $T \in \{1, 2, 3\}$.

Proof. Under every admissible hold-class control the adversarial regime realizes the declining branch at each step, so the worst stock declines at unit rate and remains safe exactly through time $z_0 - 1$ — the last safe review, the first violation occurring at z_0 : deadline-viability forces $T \leq z_0 - 1$. Conversely the hold policy tracks the decline path $z(t) = z_0 - t$ branchwise, which stays admissible exactly on $[0, z_0 - 1]$ and terminates at the floor, which is safe because the safe set is closed; hence $T \leq z_0 - 1$ is viable, boundary included. The set identity $\{T : T \leq z_0 - 1\} = \{T : T \leq \sigma^*(z_0)\}$ (integer T , all z_0) — equivalently $\{T < z_0 - 1\} = \{T \leq \sigma^*(z_0)\}$ off the integers, with the boundary failure at integer $z_0 - 1$ — gives the displayed supremum. $\square$

7. Monitoring adequacy

Let $R(x)$ denote the safe actions of state x : those keeping x in $\mathcal{V}$ for one step. On the audited target, $R((1, 2)) = \{1\}$,

Table 3: The timing grid, printed in full (93 cells): viability at the deadline against $\sigma^*(z_0) = \lfloor \tau^*(z_0) \rfloor$ (the deadline count; the identity columns use the inclusive predicate, on which the floored and real-valued forms coincide). The checkmark of column T_{obs} means the cell is viable; the identity column “id.” is the predicate $T_{\text{obs}} \leq \sigma^*(z_0)$ and column “cell” the audited viability; the two agree on all 93 cells (zero mismatches; 33 viable cells). The circled cell of Figure 2 is (2.0, 1): viable, on the boundary; the strict form disagrees exactly on the boundary cells (2, 1), (3, 2), (4, 3).

z_0	σ^*	$T_{\text{obs}} = 1$		$T_{\text{obs}} = 2$		$T_{\text{obs}} = 3$	
		id.	cell	id.	cell	id.	cell
1.0	0	.	.	.	.	.	.
1.1	0	.	.	.	.	.	.
1.2	0	.	.	.	.	.	.
1.3	0	.	.	.	.	.	.
1.4	0	.	.	.	.	.	.
1.5	0	.	.	.	.	.	.
1.6	0	.	.	.	.	.	.
1.7	0	.	.	.	.	.	.
1.8	0	.	.	.	.	.	.
1.9	0	.	.	.	.	.	.
2.0	1	✓	✓	.	.	.	.
2.1	1	✓	✓	.	.	.	.
2.2	1	✓	✓	.	.	.	.
2.3	1	✓	✓	.	.	.	.
2.4	1	✓	✓	.	.	.	.
2.5	1	✓	✓	.	.	.	.
2.6	1	✓	.	.	.	.	.
2.7	1	✓	.	.	.	.	.
2.8	1	✓	.	.	.	.	.
2.9	1	✓	.	.	.	.	.
3.0	2	✓	✓	.	.	.	.
3.1	2	✓	✓	.	.	.	.
3.2	2	✓	✓	.	.	.	.
3.3	2	✓	✓	.	.	.	.
3.4	2	✓	✓	.	.	.	.
3.5	2	✓	✓	.	.	.	.
3.6	2	✓	✓	.	.	.	.
3.7	2	✓	✓	.	.	.	.
3.8	2	✓	✓	.	.	.	.
3.9	2	✓	✓	.	.	.	.
4.0	3	✓	✓	✓	.	.	.

$R((2, 1)) = \{2\}$, $R((2, 2)) = \{1, 2\}$, and $R((3, 1)) = \{2\}$. A partition of a target belief is *adequate* when every fibre has a nonempty common safe-action set. On the four-state target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$, 7 of the 15 partitions are adequate, and two minimal adequate partitions with two cells exist. Table 4 carries the complete census with each fibre’s common safe-action set; Figure 3 displays it.

The maximal subsets with a nonempty common safe-action set are $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$. They overlap in (2, 2), so no partition into maximal common-action sets exists: coarsening to maximal fibres is not an available design, and the adequate partitions found by the census are the design options. The criterion is one-step monitoring adequacy; dynamic viability is not claimed from it (the kernels are computed by the full recursion). The census excludes instrument 0 without loss: $F_0 \leq F_1$ and $F_0 \leq F_2$ pointwise with the safe set an up-set (verified in the frame section) give every one-step-safe fibre a safe instrument in $\{1, 2\}$. Two consequences follow. Pairwise intersection of safe-action sets is necessary for fibre adequacy; in the two-instrument audited universe it is also sufficient (Helly number two: a pairwise-intersecting family of subsets of a two-point ac-

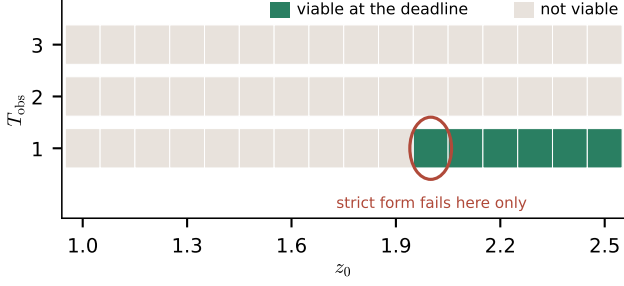


Figure 2: The timing grid. Cells viable at the deadline (green) satisfy $T_{\text{obs}} \leq \tau^*(z_0)$; the identity is boundary-inclusive, and the strict form $T_{\text{obs}} < \tau^*(z_0)$ fails at exactly one cell, $(z_0, T_{\text{obs}}) = (2.0, 1)$ (circled); against the floored count $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$ the strict form fails on all six viable cells, which is why the inclusive predicate is the audited identity.

Table 4: The fifteen-partition census on the target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$ (states written $z_1 z_2$; fibres separated by “;”). A partition is adequate when every fibre’s common safe-action set (brackets; $\cdot$ = empty) is nonempty. Seven partitions are adequate; the minimal adequate two-cell partitions are rows 6 and 7.

Partition	Adequate	Common safe actions
12 21 22 31	no	$[\cdot]$
12 21 31 ; 22	no	$[\cdot \ 12]$
12 22 31 ; 21	no	$[\cdot \ 2]$
12 31 ; 21 22	no	$[\cdot \ 2]$
12 31 ; 22 ; 21	no	$[\cdot \ 12 \ 2]$
21 22 31 ; 12	yes	$[2 \ 1]$
21 31 ; 12 22	yes	$[2 \ 1]$
21 31 ; 22 ; 12	yes	$[2 \ 12 \ 1]$
22 31 ; 12 21	no	$[2 \cdot]$
22 31 ; 21 ; 12	yes	$[2 \ 2 \ 1]$
31 ; 12 21 22	no	$[2 \cdot]$
31 ; 12 22 ; 21	yes	$[2 \ 1 \ 2]$
31 ; 21 22 ; 12	yes	$[2 \ 2 \ 1]$
31 ; 22 ; 12 21	no	$[2 \ 12 \cdot]$
31 ; 22 ; 21 ; 12	yes	$[2 \ 12 \ 2 \ 1]$

tion set has a common action), so every inadequate fibre in the census fails with a literal empty intersection, while in action sets of three or more elements pairwise intersection ceases to suffice ($\{1, 2\}, \{2, 3\}, \{1, 3\}$); and an observation map merging states with disjoint safe-action sets obstructs at that information state while every branch remains individually viable — the remedy is a fibre split, not a finer threshold on the same index. The census makes both visible at the level of individual fibres: in each inadequate partition of Table 4 the failing fibre is exactly one merging $(1, 2)$ — whose safe-action set is the singleton $\{1\}$ — with a state whose safe actions exclude 1, i.e. $(2, 1)$ or $(3, 1)$, and each adequate partition merges $(1, 2)$ with neither.

8. Regime uncertainty

Two regimes alternate: an adversary chooses at the end of step $t - 1$ the regime realized at step t . The model, explicit: the controller acts one instrument $u \in \{1, 2\}$ per step (coordinate u regenerates by one, cap 3), and the acting regime deals one unit of damage to its own coordinate (floor 0); there is no other change. With the regime frozen and announced, the controller drives the state to a fixed point — acting the damaged coordinate holds it

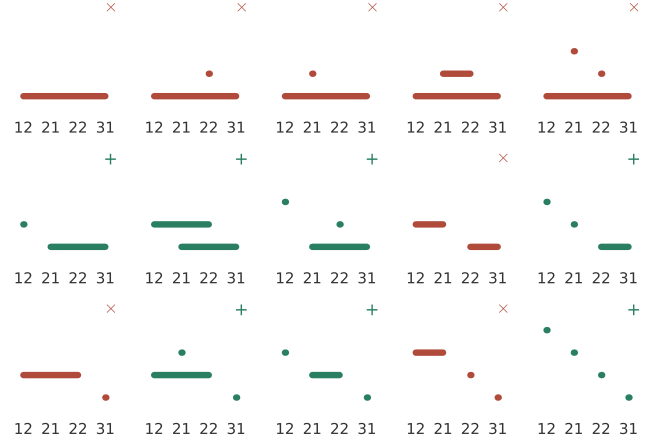


Figure 3: The census. Each panel draws one partition: a bar over a group of states is a fibre (green when its common safe-action set is nonempty, red when empty), and the margin marks adequacy (+/x). The two adequate two-cell partitions are panels 6 and 7 of the middle row.

wherever its level is below the cap, and one step of the other instrument frees a capped coordinate — so the corner $(1, 1)$ is viable under each regime separately, and each frozen kernel comprises all 36 belief pairs. Mode-blind operation, with the regime read only through the state, yields a kernel of exactly 28 — every pair containing the corner is lost (from $(1, 1)$ either instrument regenerates one coordinate while the adversarial damage zeroes the other), no merged-pair loss exists, and the loss is generated entirely at the singleton level. The mode-blind kernel agrees setwise with the full-information kernel of Section 4; the agreement is coincidental, arising under different damage routings, and no identity is claimed. All three counts (frozen 36 per regime, corner viable under each, mode-blind 28) are machine-verified from these primitives (check P13 of the edition’s script).

9. The continuous benchmark

A state is a pair of stocks $(x_1, x_2) \geq 0$ with a demand floor on each; the aggregate is $Y = x_1 + x_2$, and the Y -fibre is the set of states with that aggregate. One control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ faces the floors through Y -dependent caps $\text{cap}_1(Y) = 3/2 - (Y - 2)/10$ and $\text{cap}_2(Y) = 59/50 - (Y - 2)/10$; an allocation $u \in U$ is feasible on the fibre when $u_i \leq \text{cap}_i(Y)$ — fibre feasibility, not transition dynamics, is this section’s audited object. The critical aggregate is $Y^* = 27/5$: the cap sum equals 2 exactly there. Table 5 evaluates the caps on the audited aggregates and Figure 4 displays them. The $Y = 5$ fibre is viable with witness $(6/5, 4/5)$; the $Y = 6$ fibre is obstructed with cap sum $47/25 < 2$, certified by the dual measure $(1/2, 1/2)$ on the two active-floor atoms with margin $3/50$. The crossover is again a general law:

Proposition 6 (exact benchmark crossover). For every aggregate $Y \geq 2$ with $\text{cap}_i(Y) \geq 0$, the Y -fibre admits a feasible input (some $u \in U$ with $u_i \leq \text{cap}_i(Y)$) if and only if $\text{cap}_1(Y) + \text{cap}_2(Y) \geq 2$, which holds if and only if $Y \leq 27/5$. For $Y \geq 27/5$ the ℓ_1 -normalized dual measure $(1/2, 1/2)$ on the two cap rows certifies the obstruction with the exact margin $\frac{1}{2}(2 - \text{cap}_1(Y) - \text{cap}_2(Y)) = (Y - 27/5)/10$.

Table 5: The benchmark’s cap arithmetic on the audited aggregates (exact rationals). The cap sum crosses the demand level 2 exactly at $Y^* = 27/5$; at $Y = 6$ the cap sum falls $3/25$ short of 2, and the ℓ_1 -normalized dual measure $(1/2, 1/2)$ certifies the obstruction with margin $3/50$.

Y	cap_1	cap_2	sum	reading
4	13/10	49/50	57/25	above
9/2	5/4	93/100	109/50	above
5	6/5	22/25	52/25	viable (witness $(6/5, 4/5)$)
27/5	29/25	21/25	2	critical
11/2	23/20	83/100	99/50	below
6	11/10	39/50	47/25	obstructed, margin $3/50$

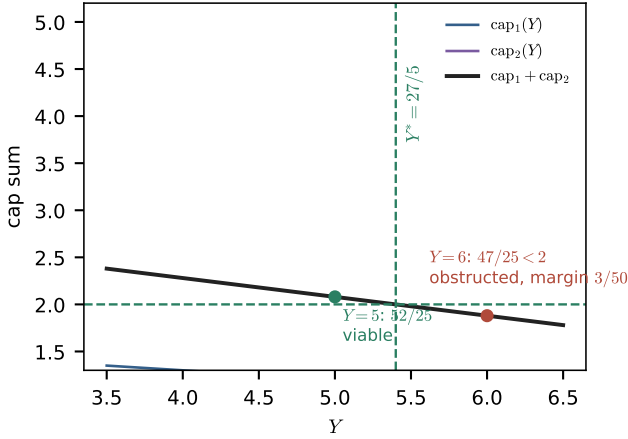


Figure 4: The benchmark. Both caps decrease linearly in the aggregate Y ; their sum crosses the demand level 2 exactly at $Y^* = 27/5$. The $Y = 5$ fibre is viable (green), the $Y = 6$ fibre obstructed with margin $3/50$ (red).

Proof. If the cap sum is at least 2, the input $u = (\text{cap}_1(Y), 2 - \text{cap}_1(Y))$ is nonnegative ($\text{cap}_1(Y) \leq 3/2 < 2$ on $Y \geq 2$), lies in U (its coordinate sum is 2), and respects the caps ($2 - \text{cap}_1(Y) \leq \text{cap}_2(Y)$ is the cap-sum inequality). Conversely any $u \in U$ under the caps has $2 \leq u_1 + u_2 \leq \text{cap}_1(Y) + \text{cap}_2(Y)$, so a cap sum below 2 is infeasible. The cap sum is the affine function $67/25 - (Y - 2)/5$, equal to 2 exactly at $Y = 27/5$ and declining beyond; the dual measure gives $\lambda^\top(\text{cap} - u) = (\text{cap sum})/2 - (u_1 + u_2)/2 \leq (\text{cap sum})/2 - 1$, the stated margin. $\square$

10. Static duality

Static duality for the benchmark’s two floor rows, abstracted to drift rows on $[0, 1]$: the max-min equals the min- λ max at value $-1/10$, with multiplier $\lambda = (1/2, 1/2)$ the ℓ_1 -normalized Farkas pair of the calculus’s worked certificate (Abaee, 2026b). Figure 5 displays the instance: under the ℓ_1 -normalized multiplier the expected drift is constant $-1/10$ on all of $[0, 1]$. Convexity is load-bearing: with $U = \{0, 1\}$ and rows $(-1, +1)$, $(+1, -1)$, the common set is empty, the max-min is -1 , and the min-multiplier maximum is 0. Under convexity the witness is sparse:

Proposition 7 (static minimax duality on the two-floor instance). On the obstructed instance with drift rows $\psi_1(u) = 2/5 - u$, $\psi_2(u) = u - 3/5$ on $U = [0, 1]$,

$$\max_{u \in [0, 1]} \min_i \psi_i(u) = \min_{\lambda \in \Delta_2} \max_{u \in [0, 1]} (\lambda_1 \psi_1 + \lambda_2 \psi_2) = -\frac{1}{10},$$

with the unique saddle point $(u^*, \lambda^*) = (1/2, (1/2, 1/2))$; under λ^* the expected drift is constant $-1/10$ on all of $[0, 1]$.

Proof. The min envelope $\min_i \psi_i(u)$ is concave and piecewise linear with its peak at the crossing $2/5 - u = u - 3/5$, i.e. $u^* = 1/2$, value $-1/10$. For $\lambda \in \Delta_2$ the pooled drift $(2\lambda_1 - 1) \cdot (-u) + \text{const}$ is affine with slope $\lambda_2 - \lambda_1 = 1 - 2\lambda_1$; it is constant exactly at $\lambda^* = (1/2, 1/2)$, where the value is $\frac{1}{2}(2/5) + \frac{1}{2}(-3/5) = -1/10$ for every u . At any other λ the maximum over $[0, 1]$ is attained at an endpoint; writing $A(\lambda)$ for the $u = 0$ endpoint value, the two endpoint values are $A(\lambda)$ and $A(1 - \lambda)$, and the averaging bound $\max(A(\lambda), A(1 - \lambda)) \geq \frac{1}{2}(A(\lambda) + A(1 - \lambda)) = -\frac{1}{10}$ — the average is exactly the constant value, for every λ — with equality only at λ^* : the convexity of the max, not symmetry alone, locates the minimum. Weak duality (max-min $\leq$ min-max) holds unconditionally; the exhibited pair attains both sides at $-1/10$, which is therefore the exact common value and the unique saddle. $\square$

Proposition 8 (sparse obstruction witnesses). An infeasible finite family of affine constraints on a convex domain in $\mathbb{R}^m$ has an infeasible subfamily of at most $m + 1$ members, and $m + 1$ is tight. The instance realizes the bound: the three constraints $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ are pairwise feasible, jointly infeasible, and no two of them certify the obstruction.

Proof. Intersect each constraint’s halfspace with the domain: an infeasible family of compact convex sets in $\mathbb{R}^m$ has an infeasible subfamily of size at most $m + 1$ (Helly’s theorem), and $m + 1$ is tight: the reversed halfspaces $\{u_i \geq 1\}$ and $\sum_i u_i \leq 1$ in $\mathbb{R}^m$ — the facets of an m -simplex, translated outward — meet m at a time and never all $m + 1$ (the facets themselves meet in the simplex, which is why the reversal is needed). For the instance, each pair is feasible — at $(1, 1)$, $(1, 0)$, and $(0, 1)$ respectively — while all three give the contradiction $2 \leq 1$; the edition’s script verifies each pair at its printed feasible point and the triple by the exact contradiction. $\square$

The Helly-tight family of three constraints $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ — no two of them already infeasible — attains the obstruction with $m + 1 = 3$ constraints. The feasible contrast at rows $(2/5 - u, u - 1/5)$ carries value $+1/10$ on both sides of the saddle: a certificate exists exactly where the value is negative, and at a zero value the instance is boundary-critical with no strictly positive margin available. Weak duality (max-min $\leq$ min-max) holds unconditionally. Strong duality holds where its hypotheses are verified: the control set compact *and* convex, the rows affine (or concave) in u , and the label family finite — as in the static instance above and the finite-horizon static-observation classes of the calculus (Abaee, 2026b). Both hypotheses are load-bearing: the paper’s own nonconvex example ($U = \{0, 1\}$, rows $(-1, +1)$, $(+1, -1)$) is compact, finite, and row-affine, yet gaps $(-1$ against $0)$; and continuity of the rows does not repair convexity, as the nonconvex rows example above shows — concavity of the rows in u , not continuity, is what the minimax argument consumes.

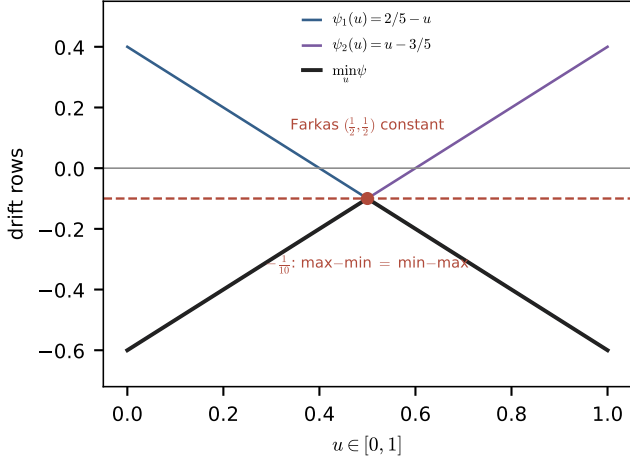


Figure 5: Static duality on the two-floor instance. The two drift rows cross at $u = 1/2$; the max–min equals the min–multiplier maximum at $-1/10$, and the ℓ_1 -normalized Farkas pair $(1/2, 1/2)$ makes the expected drift constant at the value.

Table 6: The certainty-equivalence drift audit: $g(s + 1/10) - g(s) = s/5 + 1/100$ exactly; the corrected law’s drift is 0 at every probed state.

s	uncorrected	corrected	increment per 1/4
1	21/100	0	
5/4	26/100	0	1/20
3/2	31/100	0	1/20
7/4	36/100	0	1/20
2	41/100	0	1/20

11. The certainty-equivalence drift audit

For the index $g(s) = s^2$ the uncorrected law’s one-step drift is $g(s + 1/10) - g(s) = s/5 + 1/100$ on $[1, 2]$: increasing, with minimum $21/100$ at $s = 1$ and maximum $41/100$ at $s = 2$ (Table 6); the corrected law’s drift is identically zero — by construction, the corrected law being the lead removed. A biased observation map misreads the state by a fixed one-step lead; on any decision model conditioned on the map, that misreading is the mechanism through which a nonempty kernel could empty, and the drift audit bounds it exactly — the remedy is structural correction of the map, not a better sensor.

12. Multiple floors

With several binding floors the Farkas certificate obstructs jointly: a single certificate over all floors certifies nonviability where per-floor certificates do not compose. On the audited benchmark’s two floors (Section 9), the minimal infeasible subsystem carries explicit witnesses — the Helly-tight triple of Section 10 — and the corner $(1, 1)$ is dynamically nonviable as in the shared system. *Quantifier and dual form.* Joint maintenance of several floors is the quantifier $\exists u \forall j \in I(x)$ — one control satisfying every active floor — not $\forall j \exists u_j$; per-floor certificates do not compose. The gap is visible in dimension one: on $U = [-1, 1]$ the floors $g_1(u) = u - \frac{1}{2} \geq 0$ and $g_2(u) = -u - \frac{1}{2} \geq 0$ are each separately maintainable, yet no common control maintains both, and the ℓ_1 -normalized multiplier $\lambda = (\frac{1}{2}, \frac{1}{2})$ certifies the joint infeasibility with $\sup_{u \in U} (\lambda_1 g_1 + \lambda_2 g_2)(u) = -\frac{1}{2} < 0$ — a strict-separation

dual certificate of exactly the Proposition 8 shape, with $m + 1 = 2$ constraints necessary in dimension $m = 1$ (each single floor is feasible). Scalar aggregation inherits the gap: a positive weighted average $\sum_j \lambda_j h_j$ can conceal a violated component, and a multiplier inequality $\sum_j \lambda_j D h_j \geq 0$ does not imply that every active drift is nonnegative; the ℓ_1 -normalized dual measure of Section 9 certifies the aggregate shortfall exactly because both cap rows are charged jointly. In the audited convex static examples, infeasibility is certified by a multiplier supported on at most $m + 1$ active constraints; no such bound is claimed for nonconvex controls or dynamic-policy images. Two further objects are recorded for subsequent study: a first-breach timing bound for multi-floor systems, and a corner-cone reading of the infeasibility region.

13. Design rules

The rules hold on the audited systems — each an exact census identity or a proved general law, tagged by its evidence — and transfer as default guidance only at that evidence level. *Timing:* (proved general law, Proposition 5) reviews are boundary-inclusive — $T_{\text{obs}} \leq \tau^*(z_0)$, the deadline at worst-case exit admissible under the audited endpoint convention (Section 6). *Coarseness:* (exact census, Section 7) a fibre merging states with disjoint safe-action sets obstructs at that information state; the remedy is a fibre split, not a finer threshold on the same index (Section 7). *Aggregation:* (exact census, Sections 4 and 7) the aggregate index’s fibres cross the safe boundary — each mixes states of conflicting verdict — and the aggregate kernel (26 against the full 28) certifies only a certainly-safe inner approximation; the certainly-safe set is the honest report. *Bias:* (exact audit, Section 11) the uncorrected law’s drift is at least $21/100$ on a 101-point grid of $[1, 2]$, while the corrected law’s drift is identically zero; a fixed bias is exactly correctable, and the remedy is structural correction of the map, not a better sensor. The decentralized audit adds a final rule: (proved — the restricted kernel is zero, Proposition 4) a whole-law authority restriction on one agent’s vote collapses the sustainable set outright, so protocol design precedes instrument tuning (Section 5).

14. Verification methods

Every count, witness, identity, bound, figure, and tabulated entry in this paper is regenerated by deterministic scripts in exact integer and rational arithmetic, standard library only. The master verification script recomputes the counts of Sections 3–8 from the shared system’s primitives — the transition, the fibre partition function, and the backward recursion with the no-repeat protocol threaded as a parameter — re-derives every tabulated entry of Tables 1, 2, 3, 4, 5, and 6 independently, enumerates the census of Section 7, evaluates the benchmark identities of Section 9 in rational arithmetic, asserts the drift bounds, and checks that each headline number appears in the text. The figure-and-table generation script recomputes and asserts each plotted or printed quantity from the same primitives before drawing; the two scripts agree on every shared quantity. The audited identities are additionally covered by the per-system verification scripts of the companion pa-

pers (Abaee, 2026b; see the Code availability statement).

15. Conclusion

Exact audits bind the obstruction calculus to its worked systems: each mechanism acquires a count, a witness, and a boundary. The timing certificate fires everywhere beyond $z_0 - 1$ and nowhere at the boundary; the adequate monitoring designs on the audited target are enumerable, and no coarsest one exists; policy-class value drops unequally across aggregation and protocol restrictions, with the two middle cells incomparable, the meet of their kernels institutional setwise, and exactly one pair — the alternation-forced pair — beyond both; unanimity costs a fixed and identified set of beliefs, sustained by exactly the 4×4 product of free votes, recoverable only while both votes remain free; regime blindness loses exactly the corner’s pairs, while no frozen regime loses any; and the continuous benchmark separates the viable from the obstructed at $Y^* = 27/5$ with rational certificates on both sides. The incomparability of the middle cells is a shared-register phenomenon: with per-branch registers the protocol cost collapses (Proposition 3), which is precisely why the shared register is the audited institutional object. The design rules — boundary-inclusive reviews, fibre splits over threshold tuning, certainly-safe reporting under aggregation, structural repair of bias, and protocol design before instrument tuning — are the audit’s operational content, and the master table is its evidence: every count in the paper is one column of one table, every entry of which is regenerated by script. These examples do not constitute a general taxonomy of viability failures; they supply exact counterexamples, finite censuses, and reproducible benchmarks that separate mechanisms often conflated in qualitative discussion.

References

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- Abaee, A., 2026b. An obstruction calculus for viability under incomplete observation (submitted).

Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_worked_systems_v10_verification.py` and the figure-and-table generation

script `paper2_worked_systems_figures.py` are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); they reproduce all counts, witnesses, bounds, figures, and tabulated entries verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.